

Prefab Brush+ 1.4.X Documentation

Thank you for purchasing Prefab Brush+, this documentation should get you started in using the tool to populate your environments. If you have any questions, issues or feature requests please email me at contact@archieandrews.games

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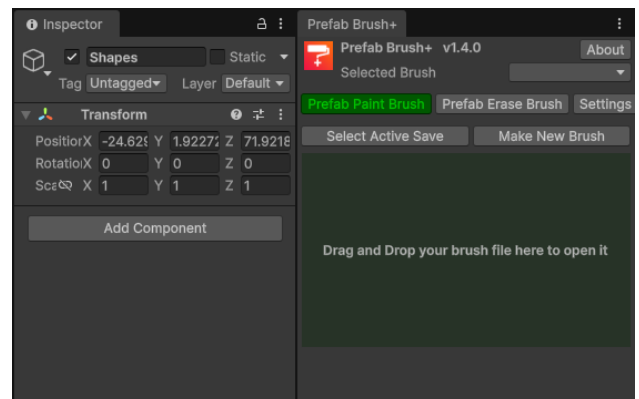
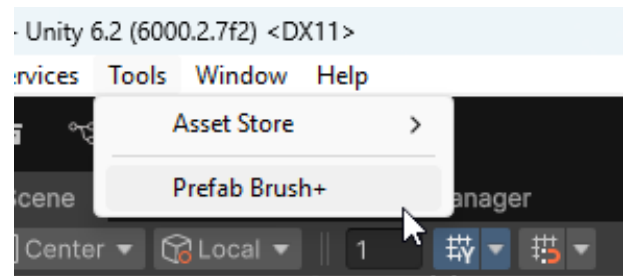
Quick Start

To open Prefab Brush+ go to the menu bar found at the top of your Unity editor and click **Tools > Prefab Brush+** to open the Prefab Brush+ Window.

Next dock the window to your preferred position in the editor. Click on the tab in the top left of the Prefab Brush window and drag it around the editor to dock. Prefab Brush+ was designed to be docked next to the Inspector window.

Prefab Brush+ uses asset files to store brush settings. You can select a compatible brush asset from the **Selected Brush** drop down menu found at the top of the Prefab Brush+ window.

You can make a new brush by pressing the **Make New Brush** button found in the top right corner of the window. This will make a new brush asset and ping it in the **Project** view.

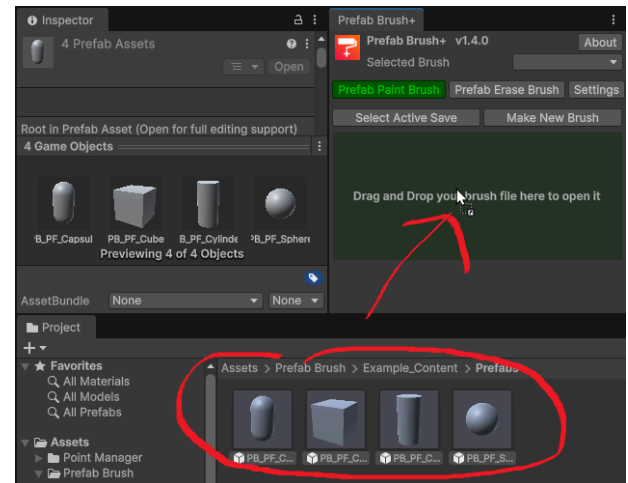
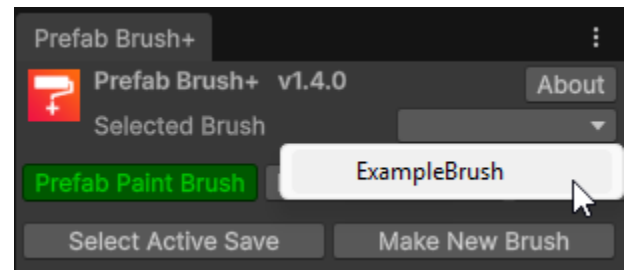


Brushes can exist in any directory in the project and Prefab Brush+ will display them in the **Selected Brush** drop down menu. When you make a new brush it should select it and show the UI for the tool. Make sure that you have selected **Prefab Paint Brush**.

To start painting prefabs we will need to add some to the brush. To do this select the prefabs you want to add from your **Project** view and drag and drop them into the green box found in the **Prefab Display** section.

Now you should be able to paint the prefabs into your scene. To use the Brush move your cursor over the scene view and left click to begin painting.

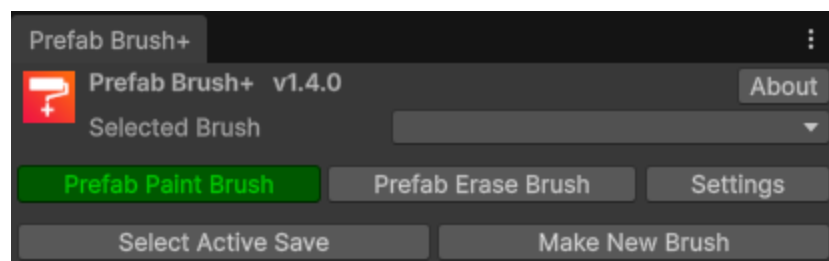
If you're having issues with the brush not painting or painted objects stacking; first double check there are colliders on the surface. You can also enable the **Paint Options > Ignore Options > Ignore Prefabs from Brush** setting to stop them stacking.



Header

At the top of the Prefab Brush+ window you can find the following...

- **Selected Brush** - This drop-down lets you select Brush assets from any directory in the **Asset** folder.
- **About** - This button shows you some basic information for the tool.
- **Prefab Paint Brush** - This button will show the prefab painting window and all of its settings.
- **Prefab Erase Brush** - This button will show the erase window and all of its settings.
- **Settings** - This button shows UI for any settings that do not fall into painting or erasing.
- **Select Active Save** - This button will ping the Brush asset you have loaded in the window in the Project View.
- **Make New Brush** - This button will create a new Brush asset and load it into Prefab Brush+.



When a Brush asset is loaded you will also see an On/Off switch to the left of the **Select Active Save** button. This will turn the brush on and off.

Brush Files

A brush file is a .asset file that stores all the information that the Prefab Brush+ window uses. In versions of Prefab Brush+ older than 1.4.0 these were called save files. The main difference between brush files and save files is that when you make a change to a setting in the PrefabBrush+ window it applies it directly to the brush file instead of needing the user to press save to apply the changes.

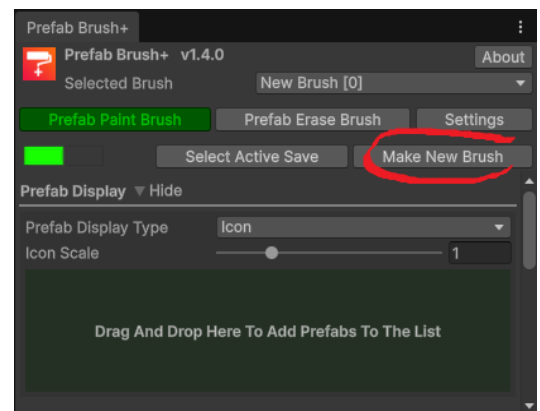


Creating Brush Files

You can create a new brush file in 2 ways...

Via The Prefab Brush+ window

When opening the Prefab Brush+ window you can find a button in the top right of the window labeled **Make New Brush**. Pressing this button will create a new brush file in the **SaveFiles** directory of the **Prefab Brush** folder.



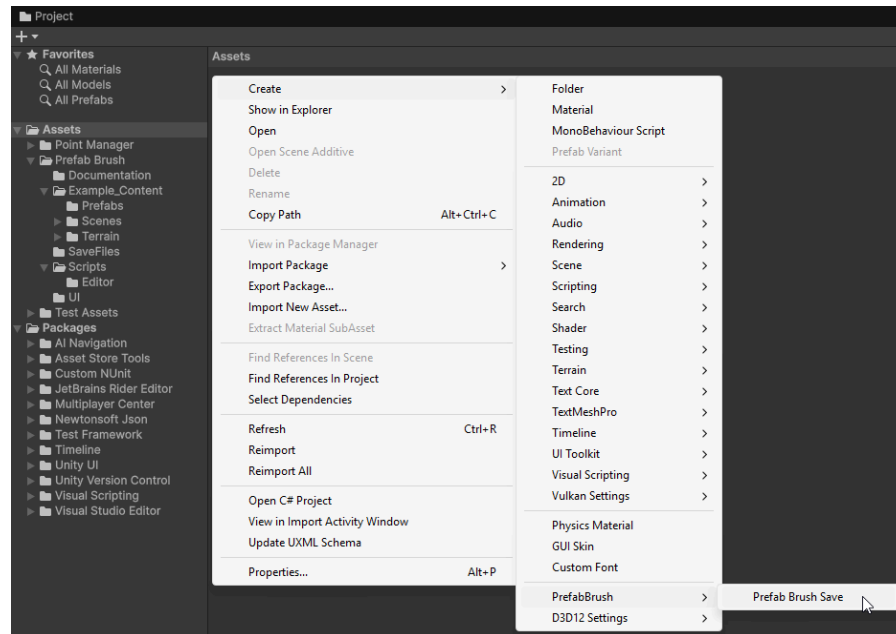
As of version 1.4.2 brush files must be in **Assets/Prefab Brush/SaveFiles** (or the relative path if you move the Prefab Brush folder)

You can add more paths for Prefab Brush to check in the Settings tab.

When you use this method to create a new brush file it will also load that brush file into the Prefab Brush+ window.

Via Unity Project Create Menu

As brush files are just scriptable objects you can also make them in the Project view window in Unity. To do so right click in any folder and go to **Create > Prefab Brush+ > Brush File**. This will create a new brush file and begin editing the name.



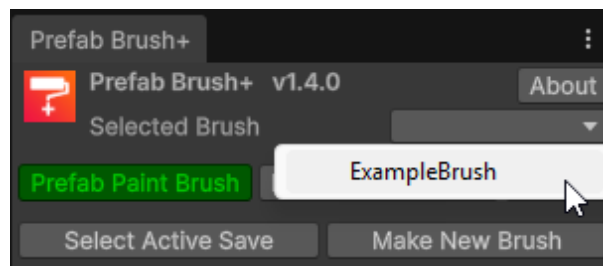
You can create brush files in any directory under the assets folder.

Loading Brush Files

Brush files can be loaded into a Prefab Brush+ window to be used. When loaded any changes made in the Prefab Brush+ window will be applied directly to the brush file. You can load a brush in 1 of 3 ways.

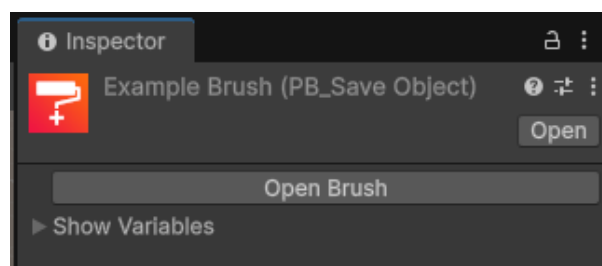
Selecting From The Drop Down Menu

At the top of the Prefab Brush+ window there is a drop down menu labeled **Selected Brush**. Clicking on the drop down menu should reveal all the brush files in your Unity project. Select the desired brush to load it into the window.



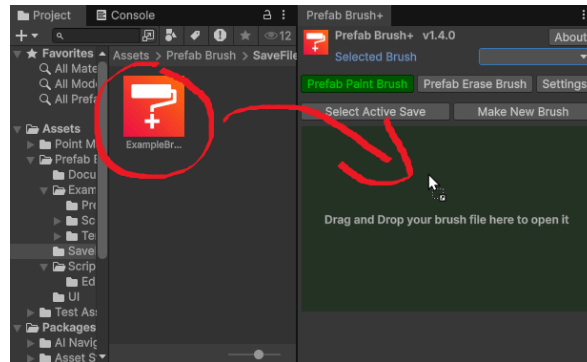
Opening From The Brush File

When selecting a brush file in the project view the **Inspector** window should show a button labeled **Open Brush**. Clicking this button will open a Prefab Brush+ window if one is not already open and load the brush file.



Dropping Into A New Prefab Brush+ Window

When you create a new Prefab Brush+ window from the Tools menu it should be blank with a large green area labeled **Drag and Drop your Brush file here to open it**. By dragging a valid brush file into this area it will be loaded into the window.



Prefab Paint Brush

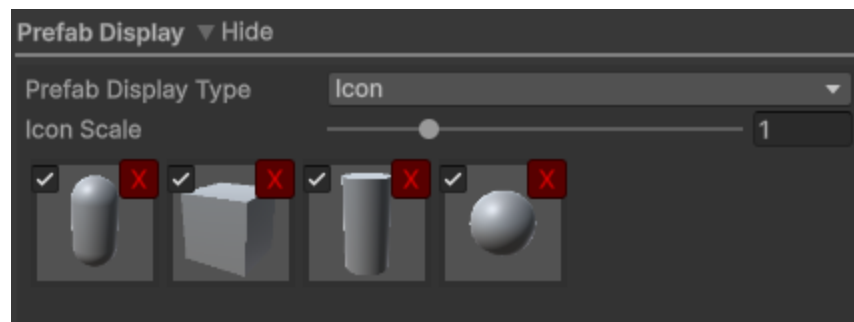
This window is where you apply settings for the prefab paint tool.

Prefab Display

Prefab Display is the area where you add and view prefabs. You can click the foldout menu button to the right of the title to show or hide the section.

Prefab Display Type

This drop down menu lets you change how Prefab Brush+ displays prefabs in the brush. You can either display them as icons or as a list. **Icon** lets you see a preview of the prefabs in the brush. **List** lets you swap out prefabs using the object fields.



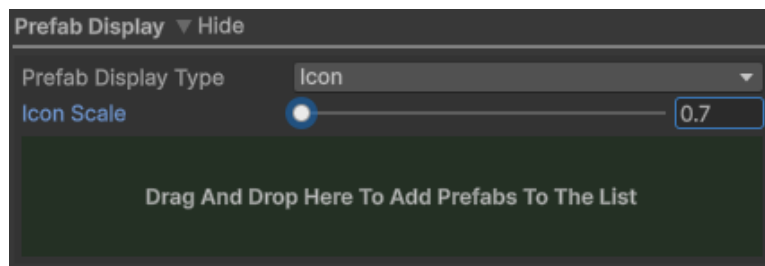
Icon Scale

When using icon mode in **Prefab Display Type** you can use the **Icon Scale** to change how big it draws the icons.



Drag and Drop Area

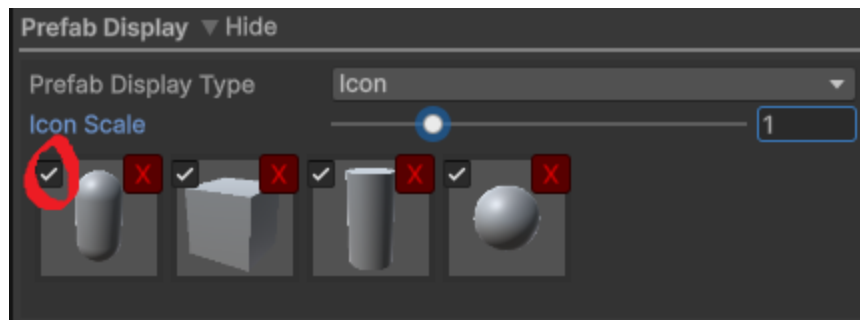
The drag and drop area is a green square that says **Drag And Drop Here To Add Prefabs To The List**. Dragging and dropping prefabs from the **Project** view to the green square will add the selected prefabs to the brush. When Prefab Display Type is set to Icon you will only see the drag and drop area when actively dragging prefabs. When using a list display a smaller drag and drop area is always visible.



Active toggle

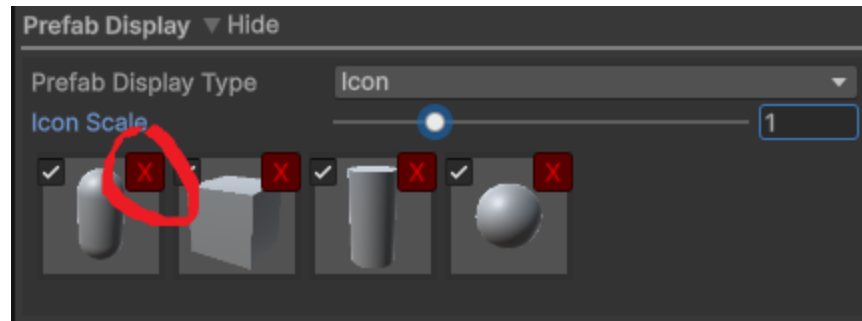
Each prefab when added to the brush will have a toggle box. This toggle box defines if the prefab is active or not in the brush.

Active means it is used in the paint brush. You can also later configure Prefab Brush+ to only erase active prefabs defined in the brush.



Remove button

To remove a prefab from the brush you need to press the red **X** button on the prefab item in the display area.



Paint Options

Under paint options you can change the type of paint tool to use and the options for ignoring some colliders as a surface.

Paint Type

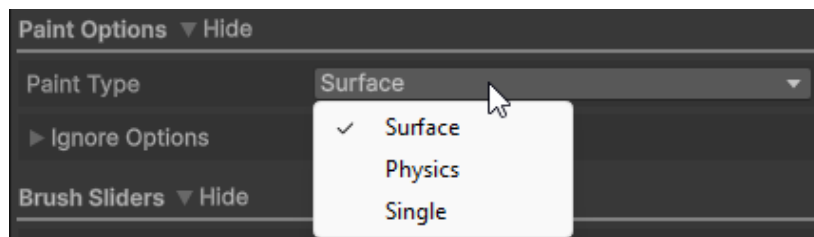
The **Paint Type** lets you set how the tool will paint prefabs onto a surface.

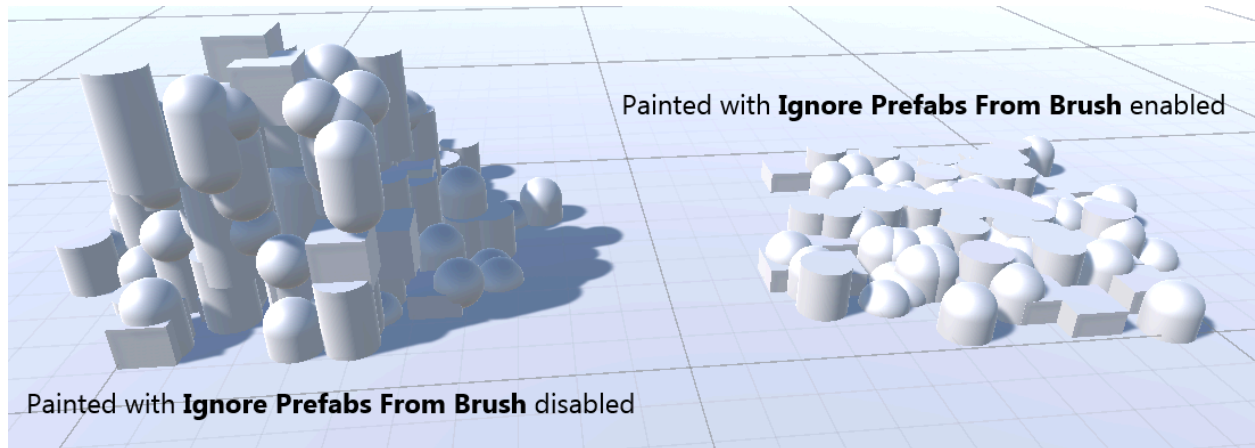
- **Surface** - Standard prefab brush, this will scatter prefabs within the painting area and apply modifiers to them once painted.
- **Physics** - Like the surface options but this will simulate physics on the painted prefab for a short period of time. There is also an option to paint the prefabs from a height so they drop down.
- **Single** - Place a single prefab at the position of your cursor and move, rotate or scale it until you release the mouse button. You can use w,e and r to change between move, rotate and scale, similar to the transform tool.

Ignore Options

By clicking on the foldout menu labeled **Ignore Options** you will see up to 3 toggle boxes.

- **Ignore Trigger Colliders** - Do not detect a trigger collider as a paintable surface.
- **Ignore Prefabs From Brush** - Do not detect the collider of a painted prefab as a surface.
- **Include Inactive Prefabs In Checks** - If using **Ignore Prefabs From Brush** this toggles if inactive prefabs are checked when painting.



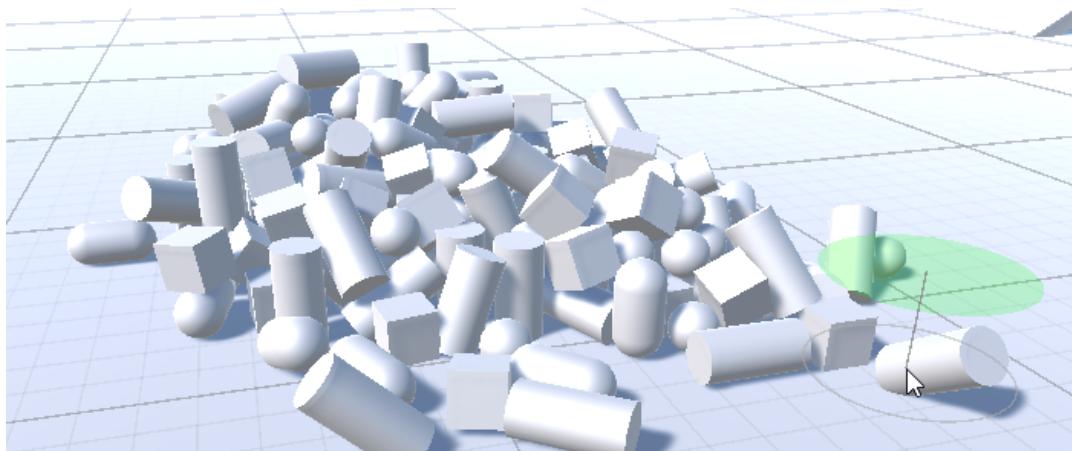


Ignoring won't prevent painting from happening but will instead treat the surface like it isn't there. In order to block painting over a specific surface look at **Filters**.

Physics Brush

When **Paint Type** is set to Physics then you will see the **Physics Brush** section. This section contains variables related to how the physics brush works.

- **Paint Height** - This sets the height offset from the brush surface that prefabs will be created.
- **Add Rigidbody To Prefab** - If your prefabs do not have Rigidbodies on them then physics simulation won't work. Enable this option to add a temporary Rigidbody component to the prefab during the duration of the simulation.
- **Physics Iterations** - How many iterations are made on each prefab after it is painted into the scene.



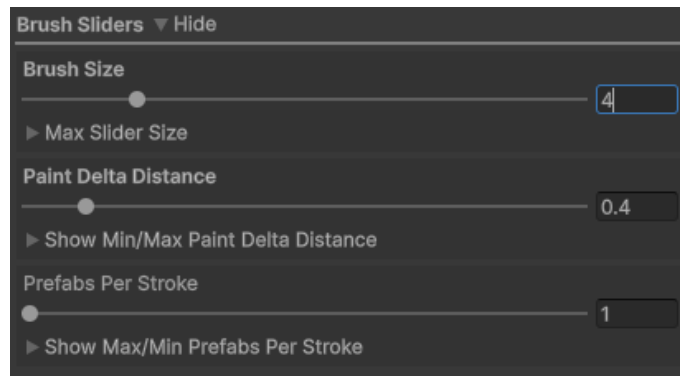
Drag Modification Type

When **Paint Type** is set to Single and the transform tool is not set to move, rotate or scale then you will see the Drag Modification Type drop down menu under the Ignore Options. This lets you change what happens after the prefab is painted and before you release the mouse button.

Swapping to move, rotate or scale on the transform tool using **W**, **E** and **R** will hide this option and match the transform tools option.

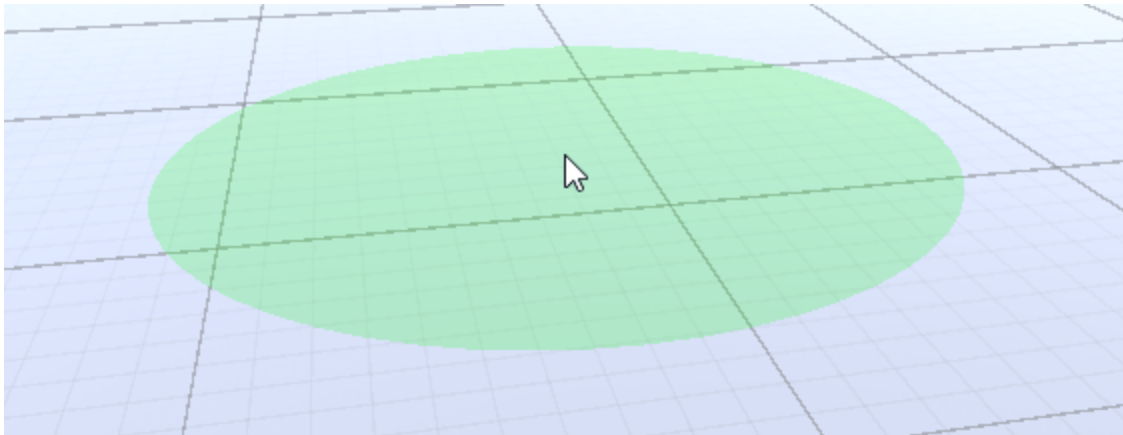
Brush Sliders

The brush slider section is home to 3 separate sliders, these sliders control the flow of prefab from the brush. Prefab Brush+ works by calling a “stroke” after a certain distance the brush travels. A stroke is when a random collection of prefabs are painted within the brushing area.



Brush Size

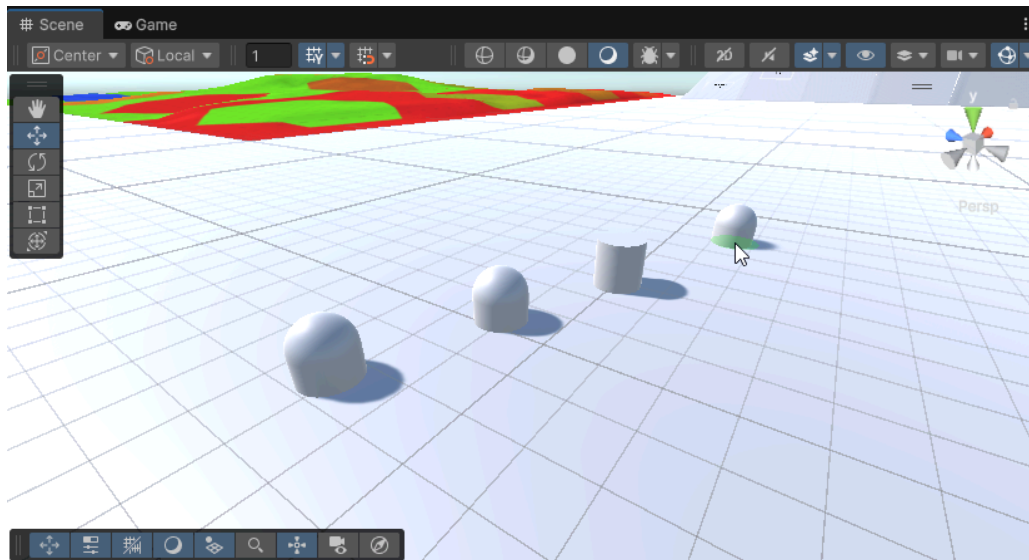
This slider changes the size of the circle in the scene view which represents the brush. Prefabs painted will be within the circle, if you want to increase the max size of the brush there is a foldout menu below the slider labeled **Max Slider Size**. Open the foldout to edit the field for the sliders max value.



Paint Delta Distance

This slider changes the distance the brush must travel in Unity units before a stroke is made. Under the slider is a foldout labeled **Show Max/Min Paint Delta Distance**, this opens 2 float fields.

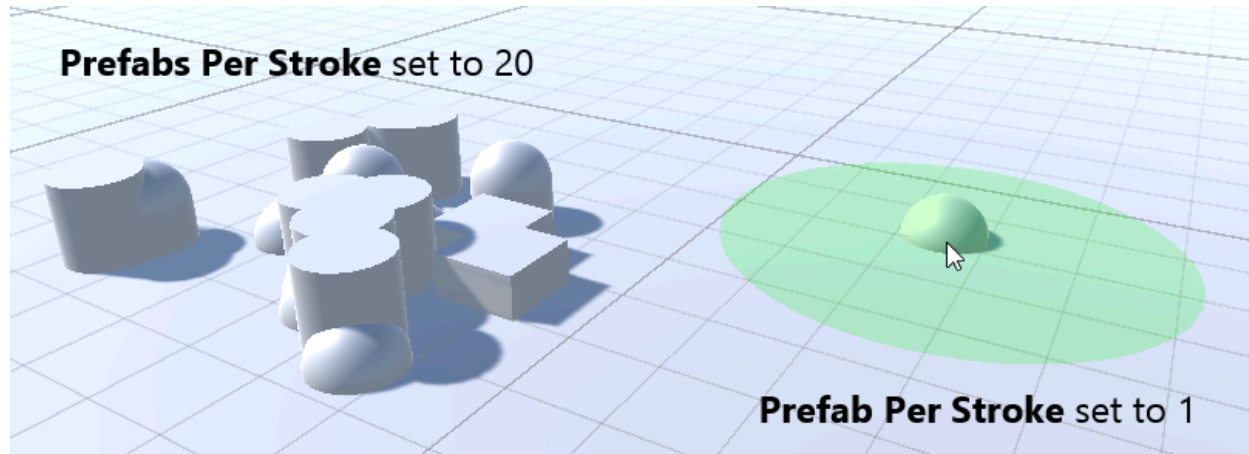
- **Min Paint Delta Distance** - The minimum value the slider can be.
- **Max Paint Delta Distance** - The maximum value the slider can be.



Prefabs Per Stroke

This slider sets how many prefabs are created within the brush area when a stroke is made. Under the slider is a foldout labeled **Show Max/Min Prefabs Per Stroke**, this opens 2 float fields.

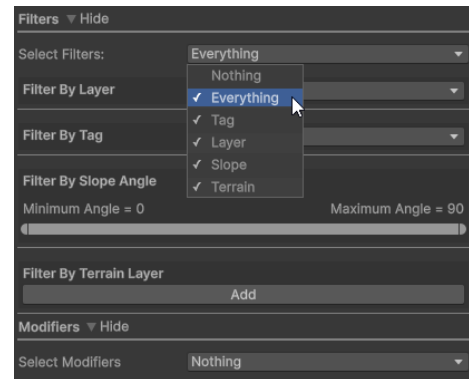
- **Min Prefab Per Stroke** - The minimum value the slider can be.
- **Max Prefab Per Stroke** - The maximum value the slider can be.



Paint Filters

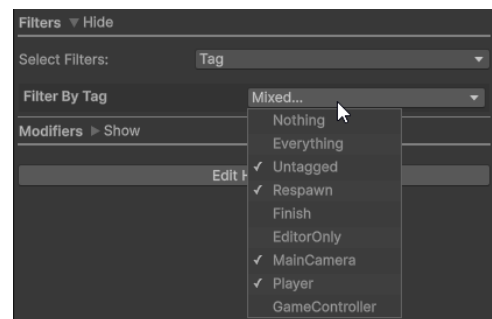
Filters are options to define what is a valid surface for the brush to paint on. Filters will check a spot in the center of the brush for a range of criteria in order to decide if it's valid. You can use filters to check surfaces using tags, layers, slopes and terrain layers.

At the top of the filters section you can find a drop down called **Select Filters:**, click on the drop down to reveal a list of filters. Select which filters you want to enable them. Any filters not selected will be inactive and the checks will not be made. Settings for filters are stored regardless if they are active or not.



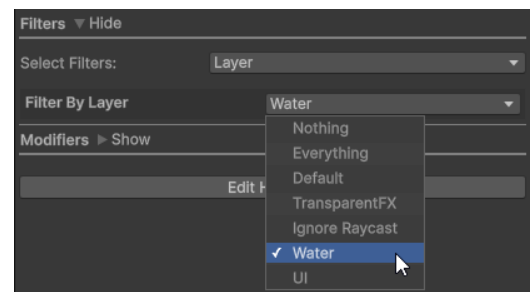
Tag

Filter by tag will let you define a tag which the surface must have in order to be counted as valid. You can select tags using the drop down menu labeled **Filter By Tag**. The surface can have any of these tags to be counted as valid. Any surfaces without the tag will not be seen as a valid surface.



Layers

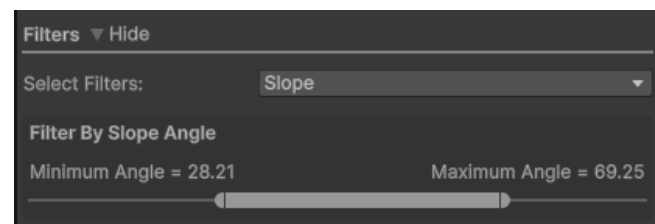
Filter by layer is the same as filter by tag but using layers instead. Using the drop down menu labeled **Filter By Layer** select the layers which you want to treat as a valid surface. The surface can have any of these layers to be counted as valid. Any surfaces without the layer will not be seen as a valid surface.



Slope

Filter by slope detects the angle sampled at the center of the brush. If the angle is within the defined limits then the surface will be counted as valid. If the slope is outside of the defined limits then the surface will not be seen as valid.

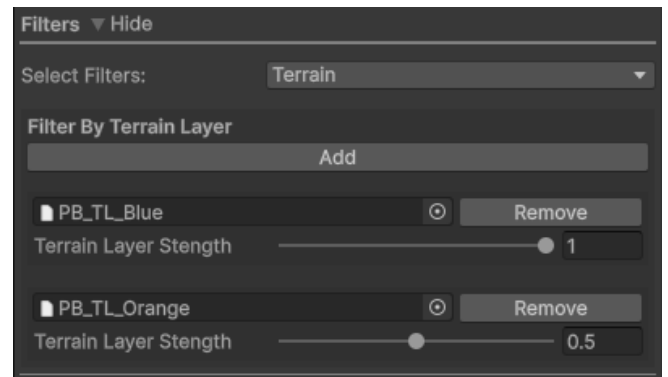
Use the Min/Max slider to define the range which is valid. A Min/Max slider has 2 handles, the left handle represents the minimum value and the right handle represents the max value. You can also click on the area between to drag both min and max points along the line.



Terrain

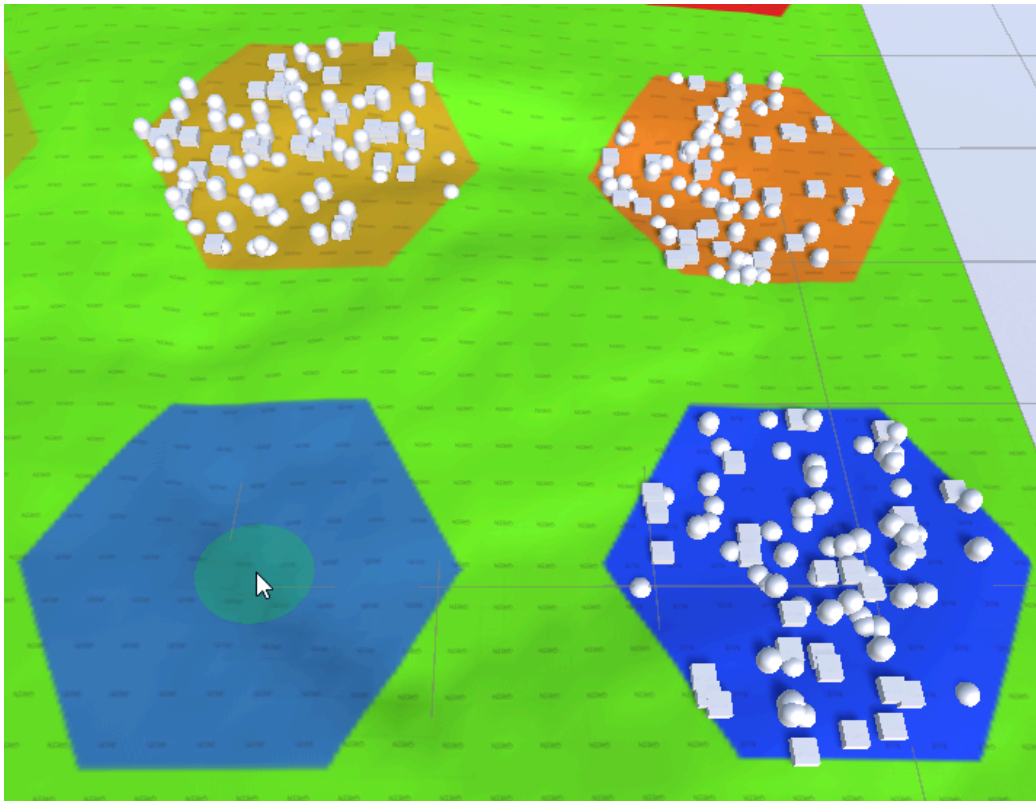
Filter by terrain layer lets you define a list of terrain layers and a strength value. When brushing over terrain Prefab Brush+ will sample the terrain layer at the center of the brush, if it matches any of the layers in the list and are above the strength value then this will be counted as a valid surface.

As terrain can have a blend of textures the strength value lets you define a minimum strength for it to count as valid. If you are not painting over terrain this filter will be ignored.



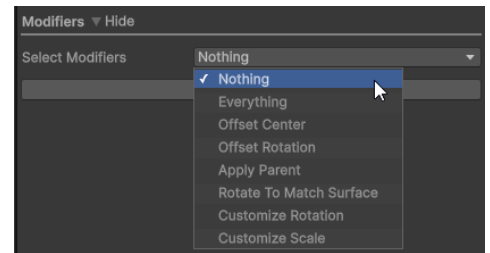
Press the **Add** button to add a new terrain layer to the list. You should see a field that you can drag and drop a **Terrain Layer** into. To remove a **Terrain Layer** from the list press the **Remove** button to the right of the **Terrain Layer** field.

Under each **Terrain Layer** item in the list there is a slider labeled **Terrain Layer Strength**. This slider goes from 0 to 1 and represents the minimum percentage of strength for that layer.



Modifiers

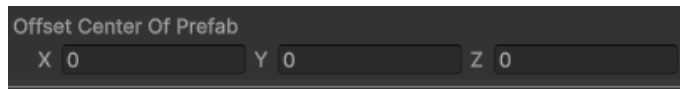
Modifiers are settings that make changes to prefabs once they are painted. This allows you to apply variation or small tweaks to prefabs as they are painted.



Offset Center

Offset center applies a Vector3 to each prefab after being painted.

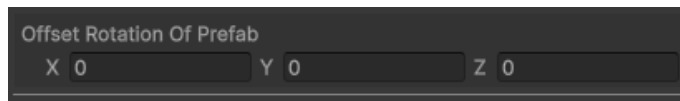
When enabled, set the offset on the X, Y and Z axis using the **Offset Center Of Prefab** field.



Offset Rotation

Offset rotation applies a Vector3 as a Euler angle to each rotation axis of the prefab after being painted.

When enabled, set the offset on the X, Y and Z axis using the **Offset Rotation Of Prefab** field.

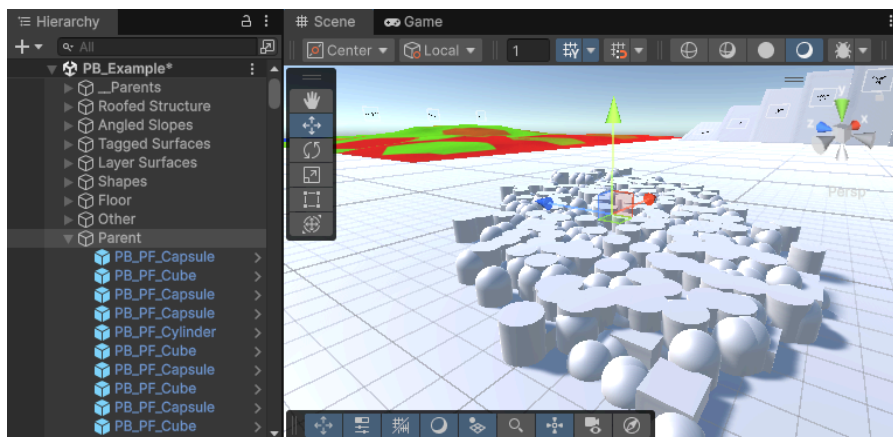


Apply Parent

The apply parent filter sets the parent of the filter using a range of options. You can change this setting using the drop down menu labeled **Parent Settings**.

- **Surface** - This will parent the painted prefab to the surface under the brush at the time of painting.
- **Single Temp Parent** - This will parent the prefab to a single GameObject defined in the brush.
- **Closest Temp From List** - This option lets you define a list of GameObjects, Prefab Brush+ will then find the closest one to the brush and parent the painted prefab to it.
- **Temp Round Robin** - This option lets you define a list of GameObjects, as you paint each prefab will be parented to one item in the list and then the next looping round.

Any GameObjects defined here will not be saved and must be set before use.

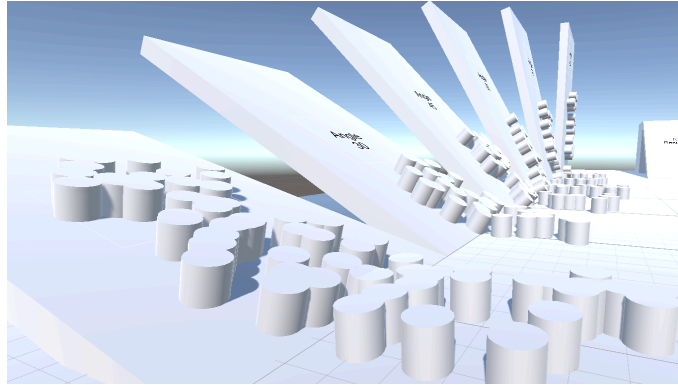


Rotate To Match Surface

Applying this filter will rotate the painted prefab so its up axis (usually y) is pointing up from the surface it is painted on.

If your prefab does not use Y as the up direction you can change it using the **Set Up Axis** drop down menu.

- **Up** - Local Y+
- **Down** - Local Y-
- **Left** - Local X-
- **Right** - Local X+
- **Forward** - Local Z+
- **Backward** - Local Z-

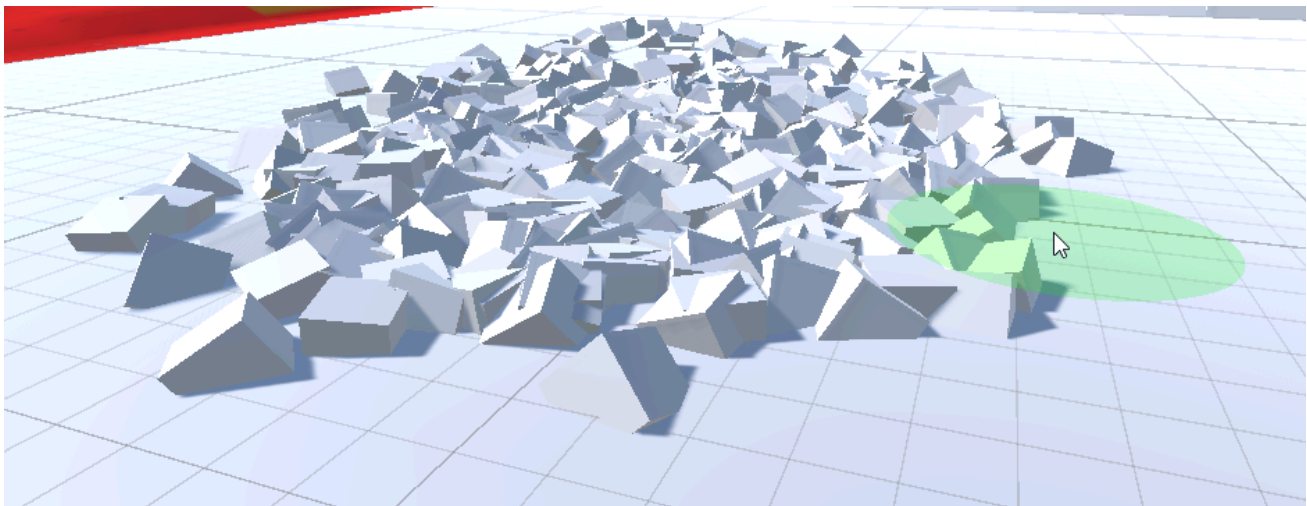


Customize Rotation

Customized rotation will apply a random rotation to each axis of the prefab after being painted.

When enabled 2 float fields will be visible for each axis. The field on the left defines the minimum value and the float on the right defines the maximum value. After painting, a random value for each axis will be created and added to the axis of the prefab.

Under the axis fields there is a button labeled **Set all to 0** pressing this button will set all fields above to match the number defined in the float field to the right labeled **Set all to value**.



Customize Scale

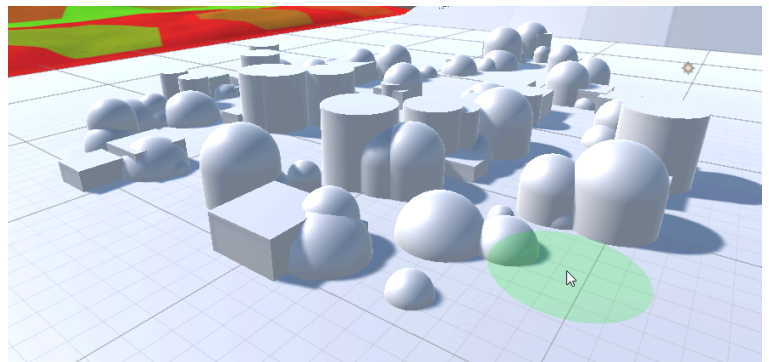
Customize scale will apply a new scale to prefabs after they have been painted. This value can be random and applied as either a set scale or a multiplication of the existing scale.

When enabled the first option is a drop down menu with 2 options labeled **Scale Type**. This option sets how the new scale is created.

- **Single Value**
 - This option will show 2 float fields one for **Minimum Scale** and one for **Maximum Scale**.
 - After painting a new scale along all 3 axis will be picked at random and applied to the prefab.
- **Multi Axis**
 - This option shows 2 float fields per axis and lets you set a minimum and maximum value per axis.
 - There is also a button labeled **Set all to 1** which when pressed sets all axis to the value defined in the **Set all to value** float field to the right.

Scale Application is the drop down menu just under the **Scale Type** drop down menu, this option sets how scale is applied to the painted prefab.

- **Set** - This will take the new value and set the scale of the prefab to match the new scale.
- **Multiply** - This will take the new scale and multiply the existing scale of the prefab by the value given.



Prefab Erase Brush

This window is where you apply settings for the prefab erase tool.

Prefab Display

This is the same as the **Prefab Display** from the **Prefab Paint Brush**. Please refer to the **Prefab Display** section found under **Prefab Paint Brush**

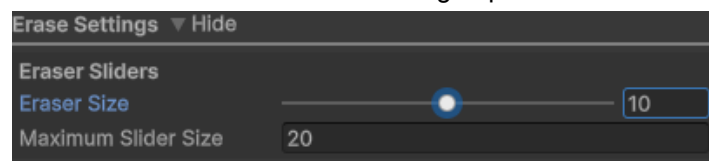
The display can be found under both as some options for the **Prefab Erase Brush** still refer to the prefabs in the brush and their active state.

Erase Settings

Eraser Sliders

The **Eraser Sliders** section has 2 options.

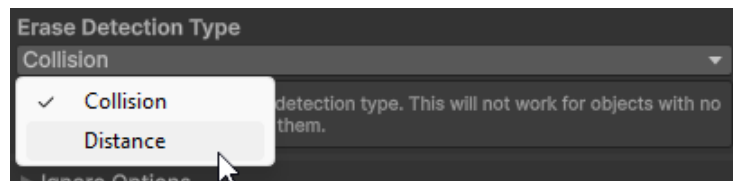
- **The Eraser Size** - The size of the erase brush in the scene view.
- **Maximum Slider Size** - The max size the slider can go up to.



Erase Detection Type

Erase detection type defines how Prefab Brush+ will detect prefabs in the scene for being erased.

- **Collision** - This will use Physics to detect objects with colliders and erase them. This option is faster and more stable.
- **Distance** - There might be some cases where you do not have a collider on the prefab you wish to erase. In these cases using the distance option will detect prefabs based on distance. This does check distance on all GameObjects in the scene. Depending on how many GameObjects you have in your scene this option can be very slow to run.



Ignore Options

Similar to the ignore option in the Prefab Paint Brush the ignore options here allow the brush to ignore certain surfaces when detecting valid surfaces. A Valid surface in erase terms means a surface that can allow an erase to happen. Please keep in mind this checks the surface under the brush and not the surface under the prefabs.

- **Ignore Trigger Colliders** - Do not detect a trigger collider as an erasable surface.
- **Ignore Prefabs From Brush** - Do not detect the collider of a painted prefab as a surface.
- **Include Inactive Prefabs In Checks** - If using Ignore **Prefabs From Brush** this toggles if inactive prefabs are checked when erasing.

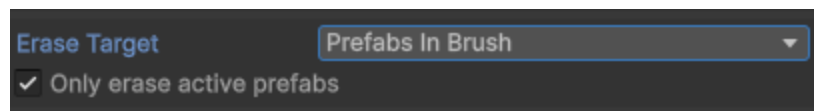
Ignoring won't prevent erasing from happening but will instead treat the surface like it isn't there. In order to block erasing over a specific surface look at **Filters**.

Erase Target

Erase Target is a drop down menu that changes how Prefab Brush+ decides if a prefab is erasable or not.

- **Prefabs In Brush** - This option will only erase prefabs that match prefabs defined in the loaded brush asset.
- **Prefabs In Bounds** - This option will detect if a bounding box of the prefab is able to fit inside of the erase area. If it does then regardless if it is a prefab defined in the brush or not it will be erased.

There is also a toggle box labeled **Only erase active prefabs**, when enabled this option prevents inactive prefabs from being erased.



Erase Filters

Filters are options to define what is a valid surface for the brush to erase on or in some cases what is a valid prefab to be erased.

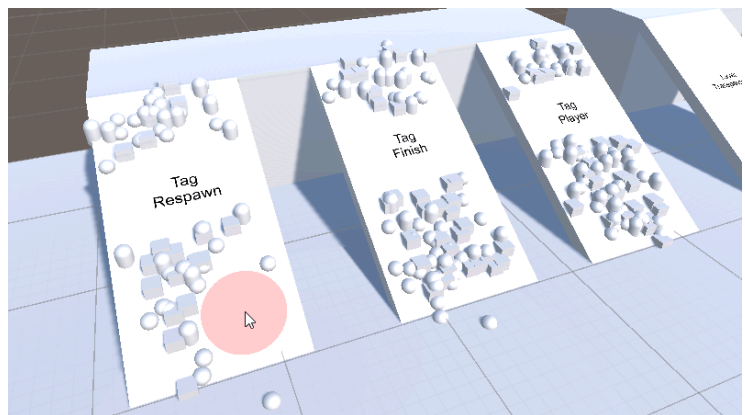
Filters will check a spot in the center of the brush for a range of criteria in order to decide if it's valid. You can use filters to check surfaces using tags, layers, slopes and terrain layers. Prefabs can be checked for just tags and layers.

At the top of the filters section you can find a drop down called **Select Filters:**, click on the drop down to reveal a list of filters. Select which filters you want to enable them. Any filters not selected will be inactive and the checks will not be made. Settings for filters are stored regardless if they are active or not.

Tag

Filter by tag will let you define a tag which the surface/prefab must have in order to be counted as valid. You can select tags using the drop down menu labeled **Filter By Tag**. The surface/prefab can have any of these tags to be counted as valid. Any surface/prefab without the tag will not be seen as a valid surface/prefab.

The drop down menu under labeled **Erase Tag Check Type** defines if you want to check the prefab or the surface for the tag.



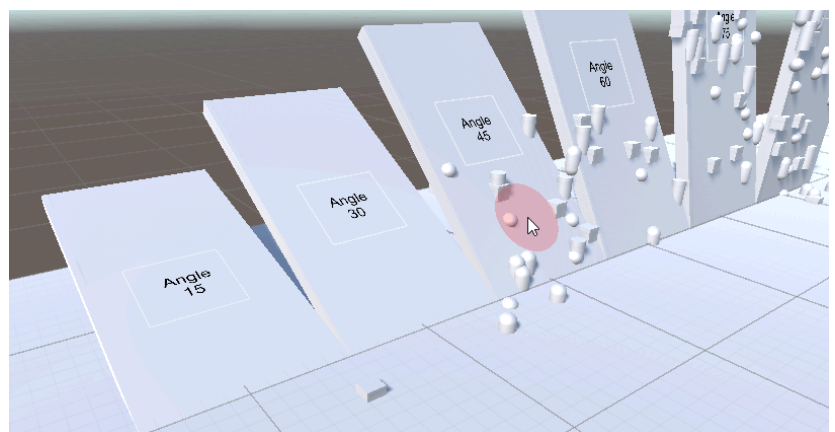
Layers

Filter by layer is the same as filter by tag but using layers instead. Using the drop down menu labeled **Filter By Layer** select the layers which you want to treat as a valid surface/prefab. The surface/prefab can have any of these layers to be counted as valid. Any surface/prefab without the layer will not be seen as a valid surface.

The drop down menu under labeled **Erase Layer Check Type** defines if you want to check the prefab or the surface for the tag.

Slope

Filter by slope detects the angle sampled at the center of the brush. If the angle is within the defined limits then the surface will be counted as valid. If the slope is outside of the defined limits then the surface will not be seen as valid.



Use the Min/Max slider to define the range which is valid. A Min/Max slider has 2 handles, the left handle represents the minimum value and the right handle represents the max value. You can also click on the area between to drag both min and max points along the line.

Terrain

Filter by terrain layer lets you define a list of terrain layers and a strength value. When brushing over terrain Prefab Brush+ will sample the terrain layer at the center of the brush, if it matches any of the layers in the list and are above the strength value then this will be counted as a valid surface.

As terrain can have a blend of textures the strength value lets you define a minimum strength for it to count as valid. If you are not erasing over terrain this filter will be ignored.

Press the **Add** button to add a new terrain layer to the list. You should see a field that you can drag and drop a **Terrain Layer** into. To remove a **Terrain Layer** from the list press the **Remove** button to the right of the **Terrain Layer** field.

Under each **Terrain Layer** item in the list there is a slider labeled **Terrain Layer Strength**. This slider goes from 0 to 1 and represents the minimum percentage of strength for that layer.

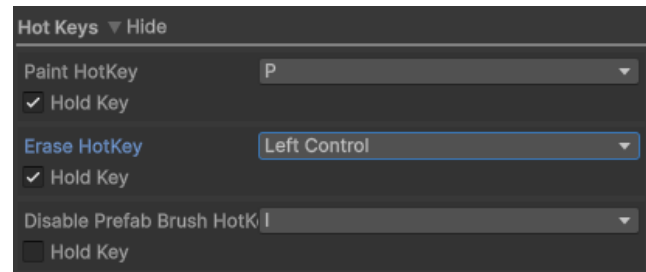
Settings

The settings view can be accessed by pressing the Settings button found at the top of the window.

Hot Keys

Three types of hotkeys can be configured by selecting a keyboard input from the following drop down menus.

- **Paint HotKey** - Switch to the Prefab Paint Brush view.
- **Erase HotKey** - Switch to the Prefab Erase Brush view.
- **Toggle Prefab Brush HotKey** - Switch the Prefab Brush+ tool on and off.



Under each drop down menu is also a toggle box labeled **Hold Key** to configure if the hotkey acts as a press or a hold.

Brush Folders

As of version 1.4.2 brush files must be in **Assets/Prefab Brush/SaveFiles**.

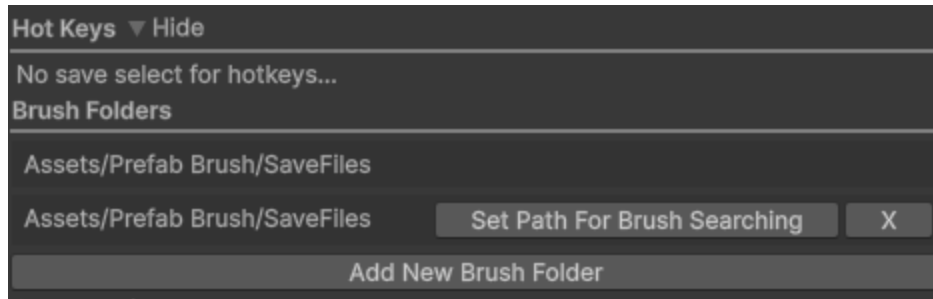
You can also add other folders for it to check in the Settings tab.

To do so click on the **Add New Brush Folder** button to create a new entry.

A new entry will be set to the default save folder. You can change it by clicking on the **Set Path For Brush Searching** button.

This will open a window, select the folder you wish to search for (It must be in the projects **Asset** folder) You can delete an entry by clicking the **X** button to the right of it.

If the folder selected in an entry is moved or deleted Prefab Brush+ will clear it from this list.



Other Settings

This area is home to any settings that do not fit under other parts of the tools UI.

The one setting here is a toggle box labeled **Show tools warning** which when enabled shows a warning message informing the user that the **Transform Tools Gizmo** has been hidden

